AudioSetCaps: An Enriched Audio-Caption Dataset Using Automated Generation Pipeline With Large Audio and Language Models

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***Abstract*—With the emergence of audio-language models, constructing large-scale paired audio-language datasets has become essential yet challenging for model development, primarily due to the time-intensive and labour-heavy demands involved. While large language models (LLMs) have improved the efficiency of synthetic audio caption generation, current approaches struggle to effectively extract and incorporate detailed audio information. In this paper, we propose an automated pipeline that integrates audio-language models for fine-grained content extraction, LLMs for synthetic caption generation, and a contrastive language-audio pretraining (CLAP) model-based refinement process to improve the quality of captions. Specifically, we employ prompt chaining techniques in the content extraction stage to obtain accurate and fine-grained audio information, while we use the refinement process to mitigate potential hallucinations in the generated captions. Leveraging the AudioSet dataset and the proposed approach, we create AudioSetCaps, a dataset comprising 1.9 million audio-caption pairs, the largest audio-caption dataset at the time of writing. The models trained with AudioSetCaps achieve state-of-the-art performance on audio-text retrieval with R@1 scores of 46.3% for text-to-audio and 59.7% for audio-to-text retrieval and automated audio captioning with the CIDEr score of 84.8. As our approach has shown promising results with AudioSetCaps, we create another dataset containing 4.1 million synthetic audio-language pairs based on the Youtube-8 M and VGGSound datasets.**

***Index Terms*—Audio-language learning, audio-language models, audio-text retrieval, automated audio captioning, large language models.**

I. Introduction

AUDIO-LANGUAGE learning plays a pivotal role in audio perception by enabling machines to process and understand relationships between audio signals and textual data [1]. This cross-modal understanding and processing ability is essential for machines to interpret, describe, and interact with the rich acoustic world in natural language. Recent advances in joint learning of audio and language representations have significantly enhanced audio understanding and reasoning capabilities [2], [3]. These advances have led to substantial progress in various applications, including speech recognition [4], [5], music composition [6], [7], and cross-modal retrieval [8], [9], moving beyond traditional audio processing methods towards more comprehensive audio understanding [10].

Audio-language models (ALMs) combine audio and language pre-trained models to enable comprehensive representation learning through audio-text pairs [11], [12], [13], [14]. A representative model is the contrastive language-audio pretraining (CLAP), which incorporates a contrastive learning framework to map audio and text into a shared multimodal space [15], [16]. This approach has demonstrated strong performance on various downstream tasks such as audio-text retrieval (ATR) [15], zero-shot emotion recognition, and zero-shot music genre classification [16]. While these advances show the potential of ALMs in audio understanding, their further development is heavily dependent on large-scale paired audio-language datasets. Constructing such datasets presents significant challenges due to the considerable time and labour investments required, creating a critical bottleneck in advancing this field [1].

To address the challenge of constructing large-scale audio-caption datasets, recent works have leveraged large language models (LLMs) for automated audio caption generation [3]. LAION-Audio-630K [15] uses sentence templates to transform web-collected tags into captions, while WavCaps [1] employs ChatGPT to refine human-annotated audio descriptions

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To facilitate research in audio-language learning, we have made our pipeline, datasets with 6 million audio-language pairs, and pre-trained models publicly available at https://github.com/JishengBai/AudioSetCaps.

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into structured captions. Although these approaches facilitate audio-caption dataset construction, they have limitations in introducing more detailed acoustic content, which has been shown beneficial for improving the quality of synthetic audio captions [17]. Auto-ACD [18] and Sound-VECaps [19] attempt to enrich acoustic caption by integrating audio-visual information. However, the requirement of paired audio-visual data increases the pipeline complexity and limits its scalability to audio-only data.

Recently proposed large audio-language models (LALMs) have shown promising abilities in extracting specific audio characteristics [2], [20], [21], [22]. LALMs advance beyond previous ALMs by integrating pre-trained audio models with LLMs for audio understanding. Pengi pioneers this direction by framing audio tasks as text generation [2], LTU enhances the framework through diverse audio question-answer (QA) pairs [20]. Qwen-Audio demonstrates strong capabilities in multi-turn dialogues and various downstream tasks by modelling audio and language information [22]. While these LALMs have achieved progress, there remains room for improvement in extracting and comprehending complex audio signals [11]. This suggests an opportunity to leverage the complementary strengths of LALMs in audio content extraction and LLMs in text generation for creating high-quality audio-language datasets.

We propose an automatic pipeline integrating LALMs, LLMs, and CLAP models to generate enriched audio-language data. This pipeline consists of three main stages: LALMs-driven audio content extraction, LLMs-assisted caption generation, and CLAP model-based caption refinement. In the first stage, we employ LALMs with a prompt chaining method to extract audio content, including overall and fine-grained audio information such as spoken language, speech emotion, music genre, and musical instruments. We then utilize LLMs to transform these audio contents into natural, detailed captions. In the final stage, we design a caption refinement module incorporating the CLAP model to filter out low-quality captions and mitigate LLM hallucinations. Using this pipeline, we generated AudioSetCaps, a dataset containing over 1.9 million (M) captions for recordings in AudioSet [23]. Subjective evaluation shows that our dataset achieves higher caption quality than existing LLMs-based datasets. We conduct experiments on multiple audio-language tasks, such as ATR, zero-shot classification, and automated audio captioning (AAC). The models trained on our dataset benefit from dataset scaling and achieve state-of-the-art performance.

Our main contributions are summarized as follows:

- We propose an automated pipeline that integrates LALMs and LLMs for audio-language dataset construction. Through prompt chaining techniques and refined caption generation, our pipeline can extract fine-grained audio content and iteratively improve the quality of captions.
- We create AudioSetCaps, a dataset comprising 1.9M audio-caption pairs. Subjective mean opinion scores indicate that the captions in AudioSetCaps are of competitive quality. Experimental results demonstrate that models trained on AudioSetCaps achieve state-of-the-art performance in audio-language tasks, surpassing the models trained on previous datasets generated using LLMs-based pipelines.
- We extend our pipeline to generate over 6 M audio-caption pairs across multiple datasets, creating the largest audio-caption dataset at the time of writing. Our pipeline, built with open-source models and tools, can run on consumer-grade GPUs, making it widely accessible. All resources related to this paper, including the proposed automatic audio labelling pipeline, datasets, and pre-trained models, are publicly available to facilitate future research in audio-language learning.

The remainder of this paper is organized as follows. Section II introduces related works in audio-language learning. Section III introduces the proposed pipeline of dataset collection and statistics of the AudioSetCaps dataset. Section IV introduces experiments on audio-language learning tasks. Section V concludes this paper.

II. Related Work

Audio-language datasets are important in advancing audio-language learning, serving as the cornerstone for developing robust ALMs. In audio captioning, current datasets can be categorized into two main types: human-annotated datasets and those generated through language model-assisted pipelines.

Audio caption datasets with human annotation form the foundation for developing and evaluating early audio captioning models. Through careful human annotation processes, experts create high-quality pairings of audio samples with descriptive text [24], [25], [26]. This approach ensures precise and contextually relevant descriptions, though it faces scalability challenges due to the labour-intensive nature of manual annotation. Two prominent datasets in this field are AudioCaps and Clotho. AudioCaps [24] is built upon AudioSet, consisting of 46,000 YouTube audio clips, focusing on 75 sound categories, where annotations were collected through Amazon Mechanical Turk[1]. AudioSet labels and corresponding video are provided as optional references during annotation to facilitate consistent and high-quality annotations. Clotho [25] contains 4,981 carefully curated audio samples from the Freesound platform, where each sample is annotated with five captions collected through workers on Amazon Mechanical Turk. The annotation process of the Clotho dataset only has access to the audio content without any additional context or hints. This makes sure the annotation is purely audio-dependent and captures different human perceptions of sounds.

Audio caption datasets with language model-assisted generation pipelines aim to address the time-consuming and labour-intensive challenge associated with human annotation. By leveraging advanced language models, these generation pipelines can efficiently construct large-scale audio-text paired datasets. In this case, the annotation pipelines typically involve pre-trained language models to generate captions based on existing audio metadata, followed by filtering and other quality control mechanisms.

[1] [Online]. Available: https://requester.mturk.com/

Several notable datasets have been developed using language model-assisted generation pipelines. LAION-Audio-630K [15], containing 633,526 pairs, uses a pre-trained T5 model to generate captions for samples with only tags or label annotations. WavCaps [1], comprising approximately 400,000 audio-caption pairs, employs a three-stage processing pipeline with ChatGPT to filter and transform web-crawled descriptions into high-quality captions. Auto-ACD [18], with over 1.5 million audio-text pairs, utilizes pre-trained models and APIs to extract audio and visual clues, then uses an LLM to organize various metadata into captions. Sound-VECaps [19] leverages an LLM to transform visual captions, audio captions, and tagging labels into descriptions, resulting in 1.66 million synthetic audio-caption pairs. LAION-Audio-630 K and WavCaps focus on transforming existing annotations but have limited fine-grained acoustic details. Auto-ACD and Sound-VECaps not only increase the pipeline complexity but also require paired audio-visual data, limiting the scalability of caption generation.

III. AudioSetCaps Pipeline and Dataset

In this section, we present a comprehensive framework for automated audio caption generation and the resulting AudioSetCaps dataset. We first introduce the large models employed in our pipeline, explaining their capabilities and roles. We then detail our three-stage pipeline, describing how each stage works together to generate enriched audio captions. Following this, we analyze the AudioSetCaps dataset in terms of its statistical characteristics and content distribution. Finally, we validate the quality of our generated captions through human evaluation.

A. Large Audio and Language Models in Pipeline

Our pipeline integrates three types of models, each selected for its specific strengths. The LALMs are used for content extraction, the LLMs handle text generation, and the CLAP model refines caption quality. This section introduces these models and describes their roles in our pipeline.

Qwen-Audio [22] is an audio understanding LALM that supports various audio tasks. Building upon Qwen-Audio, Qwen-Audio-Chat [22] can follow instructions, enable multi-turn dialogues, and support different downstream audio tasks. Qwen-Audio sets a new benchmark across multiple audio tasks, achieving state-of-the-art performance on speech recognition, vocal sound classification, and acoustic scene classification [22]. We use text prompts and audio recordings as inputs for Qwen-Audio-Chat, which generates descriptions of the overall audio content and detailed information about speech, and music.

Mistral 7B [27] is a 7.3 billion parameter language model that demonstrates good performance across various benchmarks, outperforming larger models like Llama 2 13B [28] and Llama 1 34B [29] in many areas. Mistral 7B employs grouped-query and sliding window attention methods to improve the inference efficiency of long sequences, facilitating its deployment across various tasks. In our workflow, Mistral 7B transforms the extracted audio content into complete audio captions.

CLAP [15], [16] models incorporate a contrastive learning framework to bring audio and text descriptions into a joint multimodal space, learning the mapping relationship between the two modalities. Trained on LAION-Audio-630 K, a large-scale audio caption dataset of 633,526 audio-text pairs, the LAION CLAP model [15] achieves state-of-the-art performance in text-to-audio retrieval and zero-shot audio classification. CLAP models have been used to measure the semantic similarity between audio and text in many audio tasks [7], [30]. Accordingly, we employ the LAION CLAP model [15] to evaluate the similarity between the generated captions and the corresponding audio recordings.

B. Automated Audio-Caption Generation Pipeline

The proposed automated pipeline comprises three key stages, the overview of the proposed pipeline is shown in Fig. 1. Our pipeline begins with LALMs-driven audio content extraction, where we utilize LALMs to comprehensively analyze and extract different audio metadata from the input recordings. The extracted content feeds into the LLMs-assisted caption generation stage, where LLMs transform the detailed audio information into audio captions. Finally, the CLAP model-based caption refinement stage filters and refines the captions generated by LLMs, further improving the quality of the captions. This sequential approach allows us to take advantage of each model for efficient audio caption generation.

1) LALMs-Driven Audio Content Extraction: The primary goal of LALMs-driven audio content extraction is to obtain various fine-grained audio content from audio recordings. This approach extracts detailed information about speech, music, and environmental sounds to facilitate the audio-language learning of different acoustic representations. Using this enriched content is designed to enhance the learning of audio-language relationships and improve the performance of ALMs on downstream audio tasks, such as audio captioning [17]. However, direct audio content extraction using LALMs often results in incomplete or inconsistent information, especially for complex audio scenes containing multiple elements.

To address this challenge, we employ Qwen-Audio-Chat [22], a multi-task pre-trained LALM, in combination with a prompt chaining technique. Prompt chaining [31] is designed to break down complex tasks into smaller, more manageable sub-tasks, enabling the model to handle complex tasks, without requiring additional training or fine-tuning. It has been applied in areas such as information extraction [32] and text summarization [33]. Generally, our audio content extraction process consists of three main steps:

- *Overall Audio Content Description:* Prompting Qwen-Audio to provide a concise overview of the audio, focusing on essential elements and their relationships.
- *Speech Features Extraction:* For detected speech, analyzing specific features such as emotion, gender, and language of the speaker.
- *Music Content Extraction:* For identified music, extracting information about genres and instruments.

This prompt chaining approach, adapted from LLM prompt engineering, is designed to enhance the extraction reliability of audio content by breaking down tasks into related subtasks. As shown in Fig. 2, without prompt chaining, Qwen-Audio may fail to output correct content or miss fine-grained information.

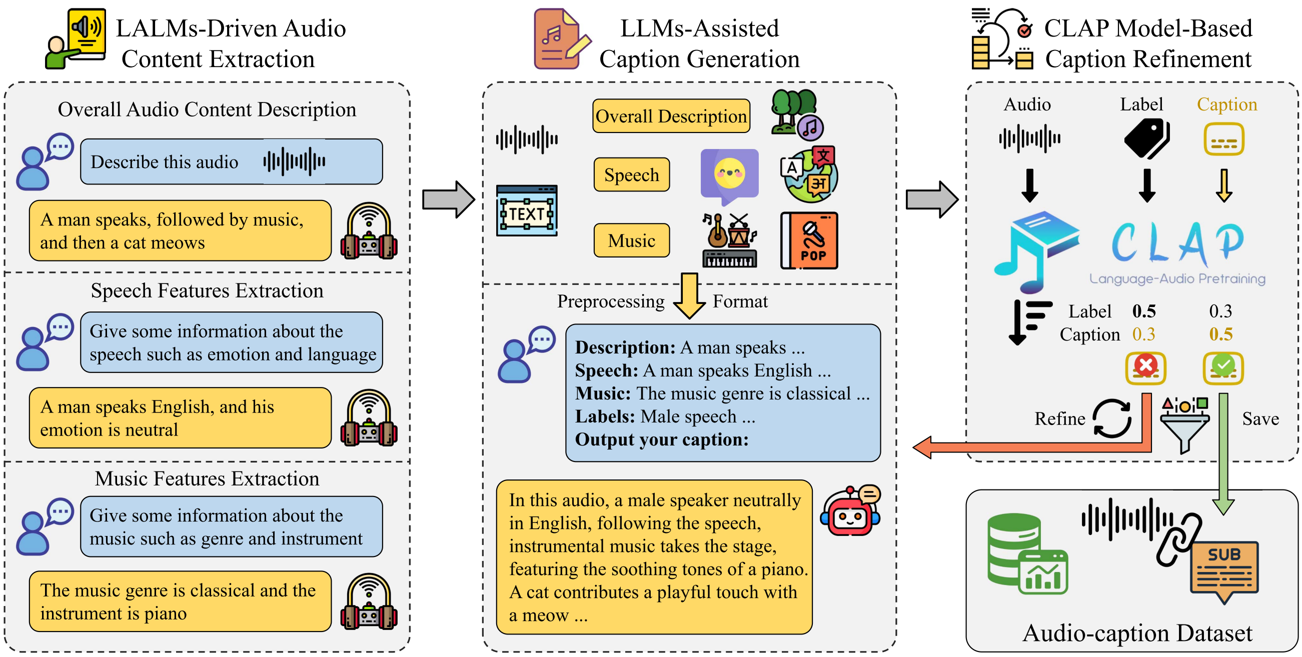

Fig. 1. Overview of the proposed automated caption generation pipeline, which integrates large audio-language models (LALMs), large language models (LLMs), and contrastive language-audio pretraining (CLAP) model.

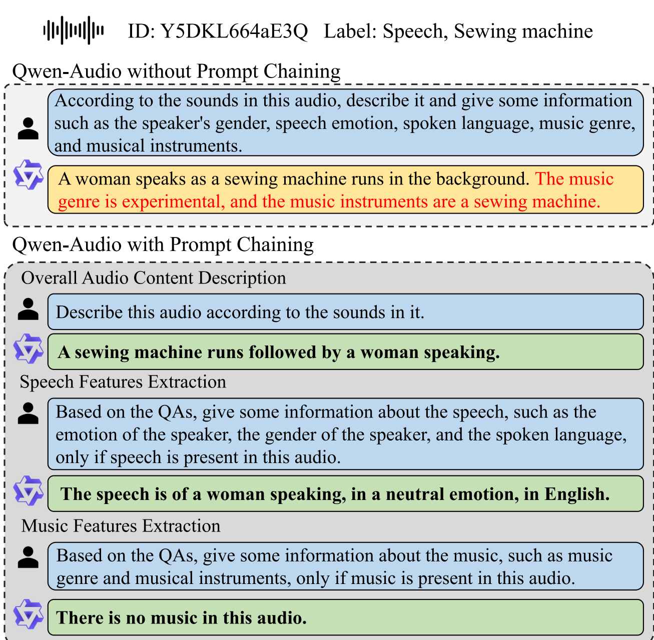

Fig. 2. An example of the output descriptions from Qwen-Audio with or without prompt chaining. Qwen-Audio without prompt chaining fails to output correct content in the second sentence (marked in red), Qwen-Audio with prompt chaining can output correct answers with fine-grained information.

2) LLMs-Assisted Caption Generation: The key challenge in caption generation is combining various types of extracted audio information while maintaining natural descriptions of audio content. In this stage, we leverage Mistral 7B [27], an LLM that can organize different audio information into audio captions using carefully designed prompts.

TABLE I
THE PROMPTS FOR LLMS-ASSISTED CAPTION GENERATION

Role Setting: I'm currently doing crowd-sourced labeling of audio data. I want you to act as a professional worker to write the final caption for the audio. The most important thing is following my instructions! Now, let's get started.
Input Details: 1. Crowd-sourced workers: Description: {*Overall Audio Content Description*} Speech: {*Speech Features Description*} Music: {*Music Features Description*} 2. The ground truth labels: {*ground truth labels*} **Instructions:** - Do not mention the specific content of speech! - Do not output the ground truth labels in the caption! - Output your caption (within 50 words):

As shown in Fig. 1, we first implement a preprocessing step that refines the LALM-extracted content using regular expression matching. This process removes phrases that indicate the absence of certain audio elements (e.g., "there is no speech/music present"), helping the LLM focus only on the audio content that is present. The refined content is then formatted into a structured prompt that integrates the information: overall audio content description, detailed speech features, and music features. To improve the reliability and accuracy of the generated captions, we include ground truth labels as references in the prompt, providing verified information about the audio content.

We design prompts with specific instructions to guide the LLM in generating appropriate captions, as shown in Table I. We prompt the LLM to play the role of a professional audio caption annotator and direct the LLM to avoid directly describing ground truth labels and mentioning specific speech recognition content in the output captions. The prompts are designed to improve the

TABLE II
AN EXAMPLE OF AN AUDIO CAPTION BEFORE AND AFTER APPLYING THE CLAP MODEL-BASED CAPTION REFINEMENT

Audio ID	YORPYVKaR0cw
Labels	Speech, Sewing machine
Before Refinement	A male voice can be heard clearly as the sound of a saw operating in the background.
After Refinement	A man is conversing while a mechanical tool, identified as a sewing machine, is operating.

The refined caption corrects the misidentified sound event from “saw” to “sewing machine”.

quality of audio captions by leveraging the language processing strengths of LLMs.

3) CLAP Model-Based Caption Refinement: Despite the strengths of LALMs and LLMs in extracting audio content and generating captions, the potential for hallucinations or inaccuracies in the generated captions remains a concern. Even with ground truth labels as a reference, LLMs may occasionally produce content that deviates from the actual audio information or includes extraneous details [34], [35]. To address this challenge, we leverage the LAION CLAP model [15], which has demonstrated strong performance in assessing audio-text alignment through its training on extensive audio-language datasets.

In the refinement stage, we implement a quality control mechanism. We first use ground truth labels as quality benchmarks by computing the similarity scores between the audio and both the generated caption and the ground truth label. When the similarity score of a caption falls below the ground truth label score, indicating potential quality issues, we trigger the regeneration process. In addition, the number of regeneration attempts can be set to a certain number to maintain computational efficiency while applying the refinement.

Table II illustrates the effectiveness of the proposed refinement process through a representative example. The initial caption incorrectly identifies a “saw” operating in the background, whereas the ground-truth label indicates a “sewing machine”. Through the CLAP-based refinement, this misidentification is corrected, resulting in a caption that more accurately describes the actual sound event.

C. Dataset Statistics and Analysis

Using the proposed pipeline, we have generated AudioSetCaps, a large-scale audio-caption dataset derived from AudioSet recordings. In this subsection, we present a comprehensive analysis of AudioSetCaps and compare it with several popular audio caption datasets. Our analysis focuses on two complementary aspects: the overall statistics that demonstrate the scale and diversity of our dataset, and the fine-grained audio content that highlights its detailed acoustic characteristics.

1) Overall Statistics: Table III presents a comparison between AudioSetCaps and several popular audio caption datasets. AudioSetCaps contains 1.9 M audio-caption pairs derived from AudioSet recordings, surpassing the scale of compared datasets. The average caption length of AudioSetCaps is 28 words, providing longer descriptions compared to LAION-Audio-630 K (7 words) and WavCaps (8 words). Although AudioSetCaps has a relatively shorter average length (28 words) and smaller vocabulary size (21 K) compared to Sound-VECaps, it focuses exclusively on audio content analysis by combining LALMs and LLMs.

TABLE III
THE STATISTICS COMPARISON OF SEVERAL AUDIO-CAPTION DATASETS

Dataset	Quan	Len	Voc	Source
Clotho [25]	30 K	11	4 K	H
AudioCaps [24]	57 K	9	5 K	H
LAION-Audio-630K [15]	630 K	7	311 K	L
WavCaps [1]	400 K	8	24 K	L
Auto-ACD [18]	1.5 M	18	20 K	A+V+L
Sound-VECaps [19]	1.6 M	40	50 K	A+V+L
AudioSetCaps	1.9 M	28	21 K	A+L

Quan denotes the number of captions, Len represents the average caption length in words, and Voc indicates the vocabulary size. Caption generation methods include human annotation (H), audio model (A), visual model (V), and language model (L).

TABLE IV
CLASSES USED TO COLLECT FINE-GRAINED AUDIO INFORMATION

Fine-grained Audio Information	**Classes**
Spoken Language	English, French, Chinese, Japanese, Arabic, Spanish, Hindi, Russian, German, Portuguese
Speech Emotion	Neutral, Calm, Angry, Excited, Sad, Happy, Fearful, Surprised, Frustrated, Nervous
Music Instrument	Guitar, Bass, Piano, Drums, Violin, Flute, Trumpet, Saxophone, Clarinet, Harp
Music Genre	Electronic, Pop, Folk, Rock, Classical, Country, Jazz, Blues, Hip hop, Reggae

TABLE V
STATISTICS OF DIFFERENT AUDIO FEATURES FOR LLMS-ASSISTED AUDIO-CAPTION DATASETS

	Language (K)	Emotion (K)	Instrument (K)	Genre (K)
WavCaps	0.01	1.4	11.4	3.8
Auto-ACD	3.5	16.6	299.8	214.3
Sound-VECaps	6.7	64.3	207.7	134.1
AudioSetCaps	13.8	336.0	690.0	682.0

Fig. 3 presents the distribution of caption lengths across different datasets, revealing distinct patterns between human annotations and automated approaches. Human-annotated datasets, i.e., Clotho and AudioCaps, exhibit concentrated distributions with most captions between 5 to 15 words, reflecting the tendency of human annotators towards concise descriptions. As for the distribution of automated approaches, WavCaps produces relatively short captions, and AudioSetCaps shows a more balanced distribution of caption length. Auto-ACD, Sound-VECaps, and AudioSetCaps generate longer descriptions than WavCaps, indicating the incorporation of other modalities in the pipeline produces longer audio captions.

2) Fine-Grained Audio Information: We count the statistics of fine-grained audio information in LLMs-based audio-language datasets to analyze the characteristics of audio-caption datasets. Table IV shows the classes for collecting fine-grained audio information, including spoken languages, speech emotions, musical instruments, and music genres. The fine-grained statistics summarized in Table V are obtained by parsing the final generated captions. AudioSetCaps reveals its comprehensive coverage of fine-grained audio content compared to other

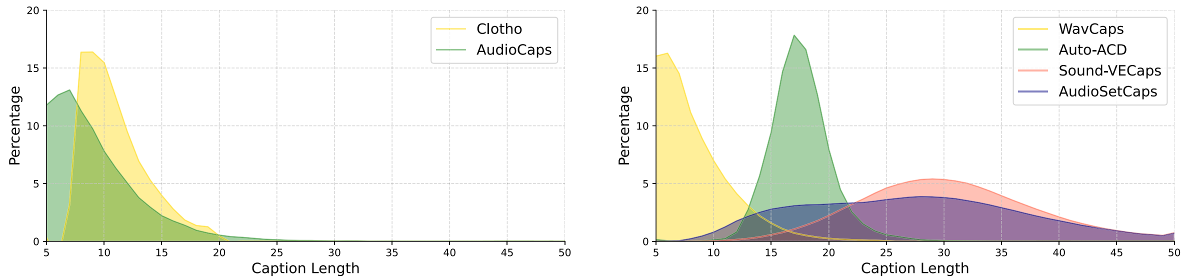

Fig. 3. Distribution of caption lengths across several popular audio caption datasets. The left subfigure shows the caption length distributions of human-labeled audio caption datasets, and the right subfigure shows the caption length distributions of LLMs-assisted audio caption datasets.

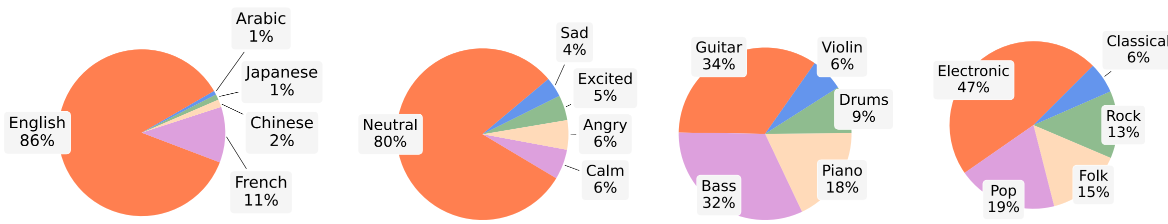

Fig. 4. Distribution of the top five categories of fine-grained speech and music information in AudioSetCaps, with pie charts representing the proportion of the most frequent categories in each type. The subfigures from left to right are pie charts of spoken languages, speech emotions, musical instruments, and music genres.

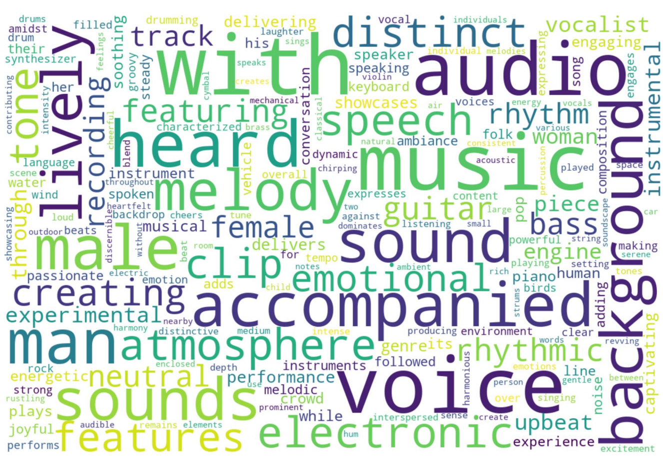

Fig. 5. Caption word cloud of AudioSetCaps excluding stop words.

LLMs-based datasets, with consistently larger numbers of identified instances across all categories. In music genre recognition, AudioSetCaps identifies 682.0 K instances, more than three times the number in Auto-ACD or Sound-VECaps.

The distribution analysis of fine-grained information in AudioSetCaps reveals distinct patterns of different categories, as illustrated in Fig. 4. In terms of languages, *English* dominates with 86% of the instances, reflecting a significant bias in the dataset. Speech emotions are dominated by *Neutral* with 80% of the instances. For musical instruments, *Guitar* and *Bass* are the most prevalent at 34% and 32%, respectively. Music genres show a notable preference for *Electronic* music at 47% .

We provide a vocabulary word cloud of AudioSetCaps to show the rich and diverse content of the dataset, shown in Fig. 5. Prominently featured are key audio elements such as "voice", "music", and "sound", indicating the comprehensive coverage of various audio types. The presence of descriptions like "female voice", "speech", and "neutral" suggests detailed annotations of speech characteristics. Musical elements are well-represented through instruments including "guitar", "bass", and "piano" and genres such as "electronic". The word cloud also reveals a focus on audio qualities and acoustic atmosphere, with descriptive terms like "distinct", "rhythmic", "emotional", and "accompanied" appearing frequently. This rich vocabulary demonstrates how AudioSetCaps captures both the acoustic content and its contextual characteristics.

D. Subjective Evaluation

To further assess the quality of AudioSetCaps, we perform subjective evaluation[2] to evaluate captions corresponding to 79 overlapping audio samples across AudioCaps [24], WavCaps [1], AudioSetCaps, and Auto-ACD [18], as well as the ground truth labels in AudioSet [23]. The question we ask for the rater is *Please listen to the provided audio samples and rate the quality of the text annotation based on its accuracy, completeness, and presence of false information.* Each caption or label is scored by at least five evaluators on how well the text annotation reflects the audio content, ranging from 1 to 5, where 1 represents "Bad", and 5 represents "Excellent".

The mean opinion score (MOS) and distributions for each dataset and the ground truth labels are presented in Fig. 6. AudioSetCaps achieves a MOS of 3.70, which closely approaches the quality of both the ground truth labels and human-labelled captions from AudioCaps. Ground truth achieves the highest MOS of 3.81, with 60.0% rated as "Good" and 12.9% as

[2] Our subjective test with Amazon Mechanical Turk has received a favourable opinion from the ethics review after completing the University of Surrey Self-Assessment Governance and Ethics form under Application No. 1046015-1045997-121266365.

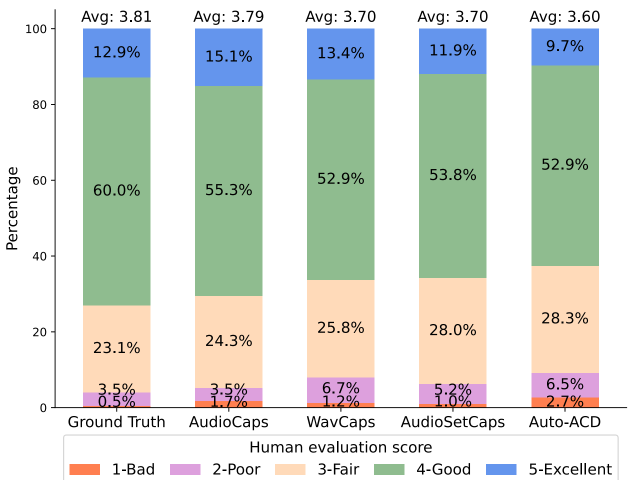

Fig. 6. Mean opinion score distribution for ground truth labels and captions of each dataset. Each score ranges from 1 (Bad) to 5 (Excellent), with the average score shown above each bar. The evaluation is conducted on 79 overlapping audio samples with at least 5 evaluators per sample.

"Excellent". AudioCaps achieves a MOS of 3.79, showing the highest percentage of "Excellent" ratings at 15.1% . WavCaps and AudioSetCaps both have a MOS of 3.70. Their score distributions are similar, where AudioSetCaps shows a slightly higher percentage of "Fair" ratings at 25.8% and a lower percentage of "Poor" ratings at 5.2% . Auto-ACD has a MOS of 3.60 with the lowest percentage of "Excellent" ratings but the highest percentage of "Fair" ratings. The automatically generated datasets, especially WavCaps and AudioSetCaps, perform comparably to human-annotated datasets.

IV. Experiments

To evaluate the proposed AudioSetCaps dataset, we conducted experiments on several audio-language tasks, including AAC, ATR, and zero-shot audio classification. Through the experiments, we aim to validate both the quality of AudioSetCaps and the effectiveness of the dataset for downstream tasks. Moreover, we provide a comprehensive analysis of the tasks including the training data scaling study, ablation study on the proposed pipeline, and a detailed case study of captions in different datasets.

A. Automated Audio Captioning

Automated audio captioning [36], [37], [38] is a cross-modal task that bridges audio understanding and natural language generation. An AAC system primarily employs an encoder-decoder neural architecture, where the encoder processes and extracts acoustic features from audio signals while the decoder generates natural language descriptions. The AAC model can be trained by a cross-entropy loss:

$$\mathcal{L}_{AAC} = -\frac{1}{N}\sum_{n=1}^{N} \log p(c_n | c_{1:n-1}, x) \tag{1}$$

where x is an input audio clip, c_n is the n-th ground truth token in a sentence with a length of N.

1) Model and Experimental Settings: The AAC model used in our experiments follows the architecture used in WavCaps [1], utilizing pre-trained HTSAT [39] as the audio encoder and BART [40] as the text decoder. For training, we pre-train the model on either AudioSetCaps alone or in combination with Clotho and AudioCaps training sets using a learning rate of 5×10^{-5} and a batch size of 24 for 15 epochs. The model is then fine-tuned on AudioCaps for 20 epochs with a learning rate of 5×10^{-6}. Note that the validation and test sets of AudioCaps are excluded during pre-training and fine-tuning. The performance is evaluated using the conventional AAC metrics including BLEU$_n$ [41], ROUGE$_l$ [42], METEOR [43], CIDEr [44], SPICE [45] and SPIDEr [46].

2) Results: Table VI presents the AAC results on the AudioCaps test set. The experimental results demonstrate the promising performance of the HTSAT and BART model trained on AudioSetCaps for AAC. The AAC model trained on the combination of AudioSetCaps, AudioCaps, and Clotho not only surpasses the performance of Qwen-Audio by a large margin but also outperforms WavCaps and many AAC methods such as Pengi [2], AL-MixGen [46], EnCLAP [47], and CoNeTTE [48]. Notably, the model trained solely on AudioSetCaps further improves the AAC performance on several metrics, outperforming the state-of-the-art AutoCap [49] across all metrics.

The comparison between models trained solely on AudioSetCaps and with combined datasets reveals an interesting pattern in performance metrics. While the model trained on combined datasets shows slightly higher performance in CIDEr and SPIDEr scores, the model trained exclusively on AudioSetCaps achieves better results in BLEU, ROUGE$_l$, METEOR, and SPICE metrics. These results suggest that AudioSetCaps contains abundant audio characteristics for caption generation, and the incorporation of additional datasets does not lead to consistent improvements across all metrics.

B. Audio-Text Retrieval

Audio-text retrieval [51], [52], [53] is a cross-modal task that aims to address the bidirectional retrieval between acoustic signals and natural language descriptions. For example, the ATR system can retrieve the most relevant text for an audio query or the best-matched audio for a text query from a candidate pool. Recent ATR systems typically adopt a contrastive learning framework, where the model learns to maximize the similarity between paired audio-text embeddings while minimizing the unpaired ones. The audio infoNCE loss [54], $\mathcal{L}_a$, can be defined as the average of a normalized function measuring the similarity of different texts to the same audio query. Similarly, the loss for the text modality, $\mathcal{L}_t$, measures the similarity of different audios to the same text query. For a batch with B audio-text pairs, we have:

$$\mathcal{L}_i^a = -\log \frac{\exp\left(e_i^a \cdot e_i^t / \tau\right)}{\sum_{j=1}^{B} \exp\left(e_i^a \cdot e_j^t / \tau\right)} \tag{2}$$

$$\mathcal{L}_i^t = -\log \frac{\exp\left(e_i^t \cdot e_i^a / \tau\right)}{\sum_{j=1}^{B} \exp\left(e_i^t \cdot e_j^a / \tau\right)} \tag{3}$$

TABLE VI
THE PERFORMANCE OF STATE-OF-THE-ART METHODS FOR AUTOMATED AUDIO CAPTIONING ON AUDIOCAPS TEST SET

Method	Training Set	Model	BLEU_1	BLEU_4	ROUGE_l	METEOR	CIDEr	SPICE	SPIDEr
Qwen-Audio [22]	N/A	Wisper+Qwen 7B	-	-	-	-	-	14.7	-
Pengi [2]	AC+CL	HTSAT+CLIP+GPT-2	69.1	25.3	48.2	23.2	75.2	18.2	46.7
AL-MixGen [46]	AC	ACT	70.0	28.9	50.2	24.2	76.9	18.1	47.5
EnCLAP [47]	AC	EnCLAP-base	-	-	-	24.7	78.0	18.6	48.3
	AC	EnCLAP-large	-	-	-	25.5	80.3	18.8	49.5
CoNeTTE [48]	AudioSet	CNeXt-Trans	-	-	-	-	-	-	49.5
	AudioSet+AC		-	-	-	-	-	-	46.6
AutoCap [49]	AC+CL+WavCaps	CLAP+HTSAT+ Q-Former+BART	72.3	29.7	51.8	25.3	83.2	18.2	50.7
WavCaps [1]	AC+CL	HTSAT+BART	67.5	27.2	48.3	23.7	71.1	17.7	44.4
	WavCaps+AC+CL		70.7	28.3	50.7	25.0	78.7	18.2	48.5
AudioSetCaps	AudioSetCaps+AC+CL	HTSAT+BART	71.4	30.3	52.2	25.7	**84.8**	18.4	**51.6**
	AudioSetCaps		**73.9**	**30.8**	**52.7**	**26.2**	83.9	**18.6**	51.3

"AC" and "CL" refer to AudioCaps and Clotho datasets. "ACT" refers to the audio captioning transformer model.

TABLE VII
PERFORMANCE OF STATE-OF-THE-ART METHODS FOR AUDIO-TEXT RETRIEVAL ON THE AUDIOCAPS TEST SET

Method	Training Set	Model	Text-to-Audio			Audio-to-Text		
			R@1	R@5	R@10	R@1	R@5	R@10
Multilingual [50]	WavCaps+AC+CL	CED-BASE+SONAR (LE)	45.9	**81.3**	**90.2**	**60.7**	**86.9**	**94.8**
LAION CLAP [15]	LA+AudioSet+AC+CL	HTSAT+RoBERTa	36.1	71.8	83.9	46.8	82.9	90.7
WavCaps [1]	AC+CL	HTSAT+BERT	39.2	74.9	86.5	49.5	81.9	91.5
	WavCaps+AC+CL	HTSAT+BERT	42.2	76.5	87.1	54.6	85.2	92.4
Auto-ACD [18]	Auto-ACD+AC+CL	HTSAT+RoBERTa (TM)	42.7	-	88.5	56.3	-	**93.9**
Sound-VECaps [19]	Sound-VECapsF	HTSAT+RoBERTa	39.2	74.1	85.0	54.0	85.5	93.2
	Sound-VECapsA	HTSAT+RoBERTa	41.2	74.5	85.3	53.3	83.2	93.0
AudioSetCaps	AudioSetCaps+AC+CL	HTSAT+BERT	43.4	78.4	88.2	57.3	84.2	93.2
		HTSAT+RoBERTa (TM)	44.8	79.2	88.8	59.6	85.1	92.2
	AudioSetCaps	HTSAT+RoBERTa	44.4	79.3	89.6	58.6	**86.2**	**93.9**
		HTSAT+RoBERTa (TM)	**46.3**	**79.7**	**90.0**	**59.7**	86.0	93.2

"LA", "AC" and "CL" refer to LAION-Audio-630K, AudioCaps and Clotho datasets. "TM" denotes the text masking method during training. "R@*k*" denotes recall at rank *k*.
Note that the multilingual method employs a multilingual language-enhanced approach, and its performance is provided here for reference only, not for direct comparison.

where τ is a temperature parameter that controls the range of logits, e_i^a and e_i^t are the embedding vectors of the i-th audio sample x_i and text caption c_i in a training batch, respectively. The loss $\mathcal{L}_{ATR}$ for the audio-text pairs in one batch can be defined as a symmetrical objective:

$$\mathcal{L}_{ATR} = \frac{1}{2B}\sum_{i=1}^{B}(\mathcal{L}_i^a + \mathcal{L}_i^t) \tag{4}$$

1) Model and Experimental Settings: We implement the ATR model following WavCaps [1], which consists of three components: a pre-trained HTSAT [39] for audio encoding, pre-trained BERT-base networks [55], [56] for text encoding, and a 2-layer multi-layer perceptron with ReLU activation for feature alignment. The model is first pre-trained either on AudioSetCaps alone or on a combined dataset of AudioSetCaps, AudioCaps, and Clotho for 40 epochs with a batch size of 196 and a learning rate of 5×10^{-5}. For supervised fine-tuning, the model is further trained on AudioCaps for 20 epochs with a learning rate of 1×10^{-5}. We evaluate the model performance using recall at rank k (R@k), which measures the percentage of queries that retrieve the correct item within the top k results.

2) Main Results: Table VII presents the ATR results on the AudioCaps test dataset. The experimental results for the ATR model trained on AudioSetCaps demonstrate state-of-the-art performance across the metrics. Compared with LAION CLAP [15], the AudioSetCaps-based models show superior results. Our primary comparison focuses on other LLM-based automated audio caption generation pipelines, such as WavCaps, Auto-ACD, and Sound-VECaps. We first train an ATR model using HTSAT and BERT on the combination of AudioSetCaps, AudioCaps, and Clotho. This model achieves an R@1 score of 43.4 for text-to-audio retrieval and an R@1 score of 57.3 for audio-to-text retrieval, surpassing the model trained on the combination of WavCaps, AudioCaps, and Clotho. Inspired by Auto-ACD [18], we introduce text masking during training to encourage stronger alignment between audio and text embeddings by guiding the model to focus on shared semantic information. When we employ HTSAT with RoBERTa [56] and the text masking method, the ATR model yields R@1 scores of 44.8 and 59.6 for text-to-audio and audio-to-text retrieval, respectively, outperforming the same ATR model used in Auto-ACD [18]. The HTSAT with RoBERTa model trained solely on AudioSetCaps attains R@1 scores of 44.4 and 58.6 for

TABLE VIII
Ablation Study of Different Pre-Training Datasets, Text Masking Ratios, and Text Encoders for Audio-Text Retrieval Models on the AudioCaps Test Set

Model	Training Set		Mask	T2A	A2T
	ASC	ASC+AC+CL	(%)	R@1	R@1
HTSAT+RoBERTa-PT	×	✓	-	40.3	54.4
	✓	×	-	30.8	41.4
HTSAT+RoBERTa-FT	×	✓	-	42.9	56.5
	✓	×	-	44.4	58.6
HTSAT+RoBERTa-FT	✓	×	25	**46.3**	**59.7**
	✓	×	50	43.4	56.4
HTSAT+BERT-FT	✓	×	-	44.8	55.7

"ASC" refers to AudioSetCaps, "AC" refers to AudioCaps, "CL" refers to Clotho, "T2A" refers text-to-audio retrieval, "A2T" refers audio-to-text retrieval, and "R@1" refers to recall at rank 1.

text-to-audio and audio-to-text retrieval, surpassing the Sound-VECaps approach. Furthermore, the HTSAT with RoBERTa model achieves improved performance when applying text masking, reaching R@1 scores of 46.3 and 59.7 for text-to-audio and audio-to-text retrieval, respectively. These results approach the performance of the state-of-the-art multilingual method [50] which uses a multilingual language-enhanced ATR framework with CED-BASE [57] and SONAR models [58].

3) Ablation Study for ATR Models: We further conduct ablation studies to analyze the impact of using different pre-training datasets, text masking ratios, and text encoders for training ATR models. Experimental results on AudioCaps test set are shown in Table VIII.

Pre-training Dataset: We study the impact of different pre-training datasets on the ATR performance using the "HTSAT+RoBERTa" model. Initially, the model is pre-trained using two configurations: solely AudioSetCaps, and a combination of AudioSetCaps, AudioCaps, and Clotho. The results show that pre-training on the combined dataset yields better initial performance. However, after fine-tuning on AudioCaps, the model pre-trained solely on AudioSetCaps achieves better performance in text-to-audio and audio-to-text retrieval tasks. This suggests that while the combined dataset can provide a better starting point, the pre-trained model solely AudioSetCaps can bridge the performance gap after the fine-tuning process on a targeted dataset like AudioCaps. Similar results can be observed from the AAC experiments that AudioSetCaps is a good audio-language learning dataset with diverse and extensive audio content.

Text Masking Ratio: We investigate the effect of different text masking ratios on the "HTSAT+RoBERTa" model pre-trained on AudioSetCaps. Three masking configurations are examined, i.e., no masking, 25% masking, and 50% masking. The experimental results show that 25% text masking effectively improves the ATR performance, with R@1 scores of 46.3% for text-to-audio retrieval and 59.7% for audio-to-text retrieval. Similarly, the ATR model in Auto-ACD also adopts 25% text masking [18]. Excessive masking, such as 50%, degrades the performance, suggesting the importance of maintaining a balance between introducing robustness and preserving semantic information during training.

Text Encoder: We study the impact of using different text encoders in ATR models. While the BERT model achieves a slightly higher R@1 score of 44.8% for text-to-audio retrieval compared to 44.4% of RoBERTa, RoBERTa demonstrates much better performance for audio-to-text retrieval. The results indicate that RoBERTa serves as a more effective text encoder for the ATR task.

C. Zero-Shot Audio Classification

Zero-shot audio classification [59], [60] is a task that can recognize unseen sound categories without further training. Leveraging the capabilities of ATR models, zero-shot audio classification generalizes to unseen classes by treating class labels or descriptions as text queries. By calculating the semantic similarity between these queries and audio embeddings within a shared embedding space, the model predicts the class with the highest similarity score as the category.

We employ the "HTSAT+RoBERTa" ATR model pre-trained on the combination of AudioSetCaps, AudioCaps, and Clotho datasets with 25% text masking, to study the zero-shot performance on audio classification tasks. We extend the labels to sentences and use them as the input text of the ATR model following the input text in WavCaps [1]. This model is evaluated on seven diverse audio classification tasks across the following three categories.

Environmental Sound: UrbanSound8K [61] (8,732 samples, 10 classes) and ESC-50 [62] (2,000 samples, 50 classes).

Speech: CommonLanguage (a subset of CommonVoice [63]) for language recognition (978 samples, 6 languages), CREMA-D [64] (7,442 samples, 6 emotions) and RAVDESS [65] (2,452 samples, 8 emotions) for emotion recognition.

Music: GTZAN-Genre [66] for genre classification (1,000 samples, 10 genres) and OpenMIC-2018 [67] for instrument recognition (13,847 samples, 20 instruments).

Results of zero-shot audio classification on environmental sound, speech, and music datasets are shown in Table IX. The ATR model trained on AudioSetCaps demonstrates strong performance across various audio classification tasks, especially on speech and music tasks. In environmental sound classification, the model achieved competitive results with 76.6% accuracy on UrbanSound8K and 88.0% on ESC-50, closely approaching the performance of WavCaps trained with diverse audio sources. For language identification using CommonLanguage, it significantly outperformed other approaches with an accuracy of 32.6% . In speech emotion recognition, the model achieved better accuracies of 28.5% on CREMA-D and 25.7% on RAVDESS. For music genre classification using GTZAN-Genre, it achieves an accuracy of 70.5%, substantially outperforming other methods. Similarly, in musical instrument recognition on OpenMIC-2018, the model attains the highest accuracy of 59.7% . These results demonstrate the effectiveness of incorporating diverse fine-grained audio content into the caption, potentially improving the generalized audio understanding for audio-language models.

D. Training Data Scaling Study

To study the impact of using different data volumes of AudioSetCaps for training, we conduct experiments using the HTSAT and BERT models on text-to-audio retrieval, audio-to-text

TABLE IX
PERFORMANCE COMPARISON OF ZERO-SHOT AUDIO CLASSIFICATION FOR ENVIRONMENTAL SOUND, SPEECH, AND MUSIC TASKS

Method	**Environmental Sound**		**Speech**			**Music**	
	Sound Classification		Spoken Language	Speech Emotion		Music Genre	Musical Instrument
	UrbanSound8K	ESC-50	CommonLanguage	CREMA-D	RAVDESS	GTZAN-Genre	OpenMIC-2018
LAION CLAP [15]	77.0	91.0	15.5	18.6	11.5	43.0	54.0
WavCaps [1]	**80.6**	**94.8**	11.8	20.1	14.5	44.3	46.2
Auto-ACD [18]	76.2	86.1	13.9	20.2	10.8	45.6	54.4
Ours	76.6	88.0	**32.6**	**28.5**	**25.7**	**70.5**	**59.7**

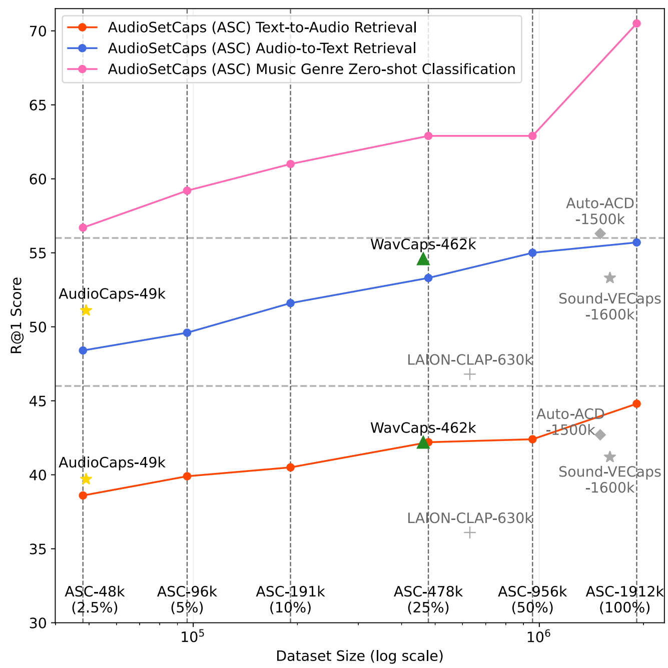

Fig. 7. Data scaling performance of AudioSetCaps and baseline methods on audio-text retrieval and zero-shot music genre classification tasks. Noted that LAION-CLAP, Auto-ACD, and Sound-VECaps are with different text encoders.

retrieval, and zero-shot music genre classification tasks. We creat six pre-defined ASC subsets: 100% (1912 k), 50% (956 k), 25% (478 k), 10% (191 k), 5% (96 k), and 2.5% (48 k), where the smaller subsets are fully included in the larger ones. By maintaining consistency across subsets, we can directly observe how model performance scales with increasing data volume. This setup also enables us to study the tradeoff between dataset scale and model performance, providing insights into the data efficiency for audio-language learning.

Fig. 7 shows the patterns in dataset scaling. For the text-to-audio retrieval task, with 96 k (5%) data from AudioSetCaps, the model surpasses the performance of AudioCaps. For the audio-to-text retrieval task, AudioSetCaps requires 191 k (10%) data to achieve comparable performance to AudioCaps. Although our approach requires more data, it offers greater efficiency and scalability. Furthermore, it ultimately matches or outperforms the performance of models trained on human-annotated data. Notably, when scaled to full size, AudioSetCaps outperforms larger datasets, including LAION-CLAP [15], WavCaps [1], Auto-ACD [18], and Sound-VECaps [19] on the text-to-audio retrieval task. For zero-shot music genre classification, we observe a trend of improvement as training data increases, with significant performance gains when scaling from 50% (956 k) to 100% (1912 k) of the dataset. These experimental results demonstrate that our proposed method exhibits promising scaling properties and benefits the performance improvement of a wide range of audio-language tasks.

TABLE X
ABLATION STUDY OF KEY COMPONENTS IN THE PROPOSED AUDIO CAPTION GENERATION PIPELINE FOR AUDIO-TEXT RETRIEVAL TASKS ON THE AUDIOCAPS TEST SET

Settings	**T2A**	**A2T**
	R@1	**R@1**
AudioCaps	39.7	51.1
AudioCaps (AudioSetCaps pipeline)	39.7	50.9
w/o Fine-grained speech and music content	38.4	48.3
w/o Prompt chaining for Qwen-Audio	39.6	49.2
Replace Qwen-Audio with LTU	36.9	47.9
w/o Caption refinement	38.7	48.9
w/ One refinement (58.2% refined)	39.7	50.7
w/ Two Refinements (41.9% refined)	39.7	50.9
w/ Three Refinements (35.3% refined)	39.6	49.7

"T2A" refers text-to-audio retrieval, "A2T" refers audio-to-text retrieval, and "R@1" refers to recall at rank 1.

E. Ablation Study on Pipeline

To evaluate the contribution of key components in the caption generation pipeline, we conduct an ablation study using the AudioCaps test set. We examine the impact of removing fine-grained speech and music content, prompt chaining for Qwen-Audio, the use of different LALMs, and the caption refinement process. We use the "HTSAT+BERT" in the experiments as the baseline ATR model while using different training sets. We train the ATR models using the AudioCaps captions and those with the same audio IDs in AudioCaps but generated by the AudioSetCaps pipeline. Experimental results are shown in Table X.

As shown in Table X, using captions generated by the AudioSetCaps pipeline yields comparable performance to human-annotated captions in AudioCaps. This indicates that the captions generated by our pipeline can serve as high-quality supervision for audio-text retrieval tasks.

We then assess the impact of removing or replacing individual components in the pipeline. Excluding the fine-grained speech and music content leads to a notable performance degradation. Next, we replace the prompt chaining by merging all the prompts for Qwen-Audio, resulting in decreased scores on the

TABLE XI
STATISTICS FOR LARGE-SCALE AUDIO-LANGUAGE DATASETS GENERATED USING THE PROPOSED PIPELINE

Dataset	Caption data	QA data	Total
AudioSetCaps	1,910,920	5,736,072	7,646,992
YouTube-8M	4,023,990	12,086,037	16,110,027
VGGSound	182,189	592,680	774,869
Total	6,117,099	18,414,789	**24,531,888**

QA denotes question-answer.

retrieval tasks. Replacing Qwen-Audio with LTU in the content extraction stage also reduces the retrieval scores, which can be attributed to Qwen-Audio's extensive multi-task pretraining and leading performance across diverse audio tasks [22]. Finally, we skip the caption refinement step during the caption generation, which also shows a decrease in the ATR performance.

We further investigate the effect of different numbers of refinement iterations. As shown in the table, about 58.2% of captions were regenerated after the first refinement, and around 41.9% after two refinements. The retrieval performance shows a notable improvement after the first refinement and achieves the best performance after the second refinement. However, the performance drop after the third refinement suggests that CLAP reaches its upper bound to provide meaningful caption evaluation and refinement.

F. Extended Audio-Language Datasets for Future Work

As demonstrated by the result shown in Tables VI, VII, and IX, and Fig. 7, AudioSetCaps significantly benefit the audio-language learning tasks and shows a strong correlation between performance gain and dataset scaling. This indicates our proposed synthetic audio caption generation pipeline works well for building large-scale audio-language datasets. To explore the boundary of data scaling and facilitate future research, we extend the proposed pipeline to generate audio-caption data for YouTube-8 M and VGGSound datasets, creating over 6 M audio captions and 18 M audio QA pairs across three datasets. The statistics of our collected audio caption and QA data are summarized in Table XI. We also make this extended audio-language dataset available to the public and will treat the experiment on this extended dataset as our future work.

V. CONCLUSION

This paper has proposed an automated pipeline that leverages large audio and language models to generate enriched audio-language data. Our pipeline comprises three key stages: LALMs-driven audio content extraction, LLMs-assisted caption generation, and CLAP model-based refinement. Using this pipeline, we created AudioSetCaps, containing 1.9 M audio-caption pairs with fine-grained speech and music contents. Extensive experiments demonstrated the effectiveness of our approach. Models trained on AudioSetCaps achieved state-of-the-art performance on AAC, ATR, and zero-shot audio classification tasks. Our ablation studies verified the importance of each pipeline component, while the data scaling analysis showed that even a subset of AudioSetCaps can provide competitive performance compared to existing datasets.

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